

Relations & Functions

- **Relation:** A subset of the Cartesian product $A \times B$.
- **Function:** A relation $f : A \rightarrow B$ such that every element of A has one and only one image in B .
 - **One-to-one (Injective):** $f(x_1) = f(x_2) \implies x_1 = x_2$.
 - **Onto (Surjective):** For every $y \in B$, there exists $x \in A$ such that $f(x) = y$.
 - **Bijective:** Both one-to-one and onto.
- **Composition of Functions:** $(g \circ f)(x) = g(f(x))$.
- **Inverse Function:** $f^{-1} : B \rightarrow A$ exists if f is bijective. $f^{-1}(y) = x \iff f(x) = y$.

Inverse Trigonometric Functions

- **Definitions:**
 - $\sin^{-1} x$: Domain $[-1, 1]$, Range $[-\pi/2, \pi/2]$
 - $\cos^{-1} x$: Domain $[-1, 1]$, Range $[0, \pi]$
 - $\tan^{-1} x$: Domain $(-\infty, \infty)$, Range $(-\pi/2, \pi/2)$
- **Properties:**
 - $\sin^{-1} x + \cos^{-1} x = \pi/2$
 - $\tan^{-1} x + \cot^{-1} x = \pi/2$
 - $\sin^{-1}(-x) = -\sin^{-1} x$
 - $\cos^{-1}(-x) = \pi - \cos^{-1} x$
 - $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$

Matrices

- **Order:** $m \times n$ (rows $\times$ columns).
- **Types:** Row, Column, Square, Diagonal, Scalar, Identity, Zero matrix.
- **Operations:**
 - **Addition/Subtraction:** Only for same order matrices. $(A+B)(ij) = A(ij) + B(ij)$.
 - **Scalar Multiplication:** $(kA)(ij) = kA(ij)$.
 - **Multiplication:** $A_{m \times p} B_{p \times n} = C_{m \times n}$. $(AB)(ij) = \sum_{k=1}^p A_{ik} B_{kj}$.
 - **Transpose:** A^T or A' . $(A^T)^T = A$. $(A+B)^T = A^T + B^T$. $(kA)^T = kA^T$. $(AB)^T = B^T A^T$.
- **Symmetric Matrix:** $A^T = A$.
- **Skew-Symmetric Matrix:** $A^T = -A$.
- Any square matrix can be expressed as sum of symmetric and skew-symmetric matrix: $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$.

Determinants

- **Definition:** A scalar value associated with a square matrix.
- **Order 2:** $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.
- **Order 3:** $\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$.
- **Properties:**
 - If any two rows/columns are identical or proportional, $\det(A) = 0$.
 - If a row/column consists of all zeros, $\det(A) = 0$.
 - $\det(A^T) = \det(A)$.
 - $\det(AB) = \det(A) \det(B)$.
 - $\det(kA) = k^n \det(A)$ for $n \times n$ matrix.
- **Minors (M_{ij}):** Determinant of submatrix obtained by deleting i -th row and j -th column.
- **Cofactors (C_{ij}):** $C_{ij} = (-1)^{i+j} M_{ij}$.
- **Adjoint of a Matrix ($\text{adj}(A)$):** Transpose of the cofactor matrix. $\text{adj}(A) = [C_{ji}]$.
- **Inverse of a Matrix (A^{-1}):** $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$, if $\det(A) \neq 0$.
- **Singular Matrix:** $\det(A) = 0$. Non-singular if $\det(A) \neq 0$.
- **Solving System of Linear Equations (Matrix Method):** $AX = B \implies X = A^{-1}B$.

Continuity & Differentiability

- **Continuity:** A function $f(x)$ is continuous at $x = c$ if $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$.
- **Differentiability:** A function $f(x)$ is differentiable at $x = c$ if $\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$ exists.
 - If a function is differentiable at a point, it is continuous at that point. The converse is not true.
- **Derivatives of Standard Functions:**
 - $\frac{d}{dx}(x^n) = nx^{n-1}$
 - $\frac{d}{dx}(\sin x) = \cos x$

- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(\log x) = 1/x$
- **Chain Rule:** $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$.
- **Implicit Differentiation:** Differentiate terms with respect to x , treating y as a function of x (use chain rule for y terms).
- **Logarithmic Differentiation:** Used for functions of the form $[f(x)]^{g(x)}$ or products/quotients of many functions. Take log on both sides.

Applications Of Derivatives

- **Rates of Change:** $\frac{dy}{dx}$ represents the rate of change of y with respect to x .
- **Increasing/Decreasing Functions:**
 - $f'(x) > 0 \implies f(x)$ is increasing.
 - $f'(x) < 0 \implies f(x)$ is decreasing.
- **Tangents and Normals:**
 - Slope of tangent at (x_0, y_0) is $m_T = f'(x_0)$.
 - Equation of tangent: $y - y_0 = m_T(x - x_0)$.
 - Slope of normal: $m_N = -1/m_T$ (if $m_T \neq 0$).
 - Equation of normal: $y - y_0 = m_N(x - x_0)$.
- **Maxima and Minima:**
 - **First Derivative Test:**
 - If $f'(x)$ changes sign from +ve to -ve at c , c is a point of local maxima.
 - If $f'(x)$ changes sign from -ve to +ve at c , c is a point of local minima.
 - **Second Derivative Test:**
 - Find $f'(x)$, set $f'(x) = 0$ to find critical points c .
 - Find $f''(x)$.
 - If $f''(c) < 0$, c is a point of local maxima.
 - If $f''(c) > 0$, c is a point of local minima.
 - If $f''(c) = 0$, test fails (use first derivative test).
- **Approximations:** $f(x + \Delta x) \approx f(x) + f'(x)\Delta x$.

Integrals

- **Indefinite Integral:** $\int f(x)dx = F(x) + C$, where $F'(x) = f(x)$.
- **Standard Formulas:**
 - $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ ($n \neq -1$)
 - $\int \frac{1}{x} dx = \log |x| + C$
 - $\int \sin x dx = -\cos x + C$
 - $\int \cos x dx = \sin x + C$
 - $\int e^x dx = e^x + C$
 - $\int a^x dx = \frac{a^x}{\log a} + C$
- **Integration by Substitution:** Let $u = g(x)$, then $du = g'(x)dx$.
- **Integration by Parts:** $\int u dv = uv - \int v du$. (LIATE rule for choosing u)
- **Partial Fractions:** For rational functions $\frac{P(x)}{Q(x)}$.
 - $\frac{px+q}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$
 - $\frac{px+q}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$
 - $\frac{px^2+qx+r}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$ (if $x^2 + bx + c$ is irreducible)
- **Definite Integral:** $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$.
- **Properties of Definite Integrals:**
 - $\int_a^b f(x)dx = \int_a^b f(t)dt$
 - $\int_a^b f(x)dx = -\int_b^a f(x)dx$
 - $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$
 - $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$
 - $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$
 - $\int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx$ if $f(x)$ is even ($f(-x) = f(x)$)
 - $\int_{-a}^a f(x)dx = 0$ if $f(x)$ is odd ($f(-x) = -f(x)$)

Applications Of The Integrals

- **Area under simple curves:**
 - Area of region bounded by $y = f(x)$, x -axis, $x = a$, $x = b$ is $\int_a^b y dx$.
 - Area of region bounded by $x = g(y)$, y -axis, $y = c$, $y = d$ is $\int_c^d x dy$.
- **Area between two curves:**
 - Area bounded by $y = f(x)$ and $y = g(x)$ (where $f(x) \geq g(x)$) from $x = a$ to $x = b$ is $\int_a^b [f(x) - g(x)]dx$.

Differential Equation

- **Order:** Highest order derivative present in the equation.

- **Degree:** Highest power of the highest order derivative (after making it polynomial in derivatives).
- **Types of DEs:**
 - **Variable Separable:** $\frac{dy}{dx} = f(x)g(y) \implies \int \frac{dy}{g(y)} = \int f(x)dx$.
 - **Homogeneous DE:** $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$. Substitute $y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$.
 - **Linear DE:** $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x or constants.
 - Integrating Factor (IF): $e^{\int P dx}$.
 - Solution: $y \cdot (\text{IF}) = \int Q \cdot (\text{IF})dx + C$.
 - Also $\frac{dy}{dx} + P'y = Q'$ where P', Q' are functions of y or constants.
 - IF: $e^{\int P' dy}$.
 - Solution: $x \cdot (\text{IF}) = \int Q' \cdot (\text{IF})dy + C$.



Vector Algebra

- **Vectors:** Quantity with magnitude and direction.
- **Scalar:** Quantity with only magnitude.
- **Position Vector:** $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.
- **Magnitude:** $|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$.
- **Unit Vector:** $\hat{r} = \frac{\vec{r}}{|\vec{r}|}$.
- **Dot Product (Scalar Product):** $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta = a_1b_1 + a_2b_2 + a_3b_3$.
 - $\vec{a} \cdot \vec{a} = |\vec{a}|^2$.
 - If $\vec{a} \cdot \vec{b} = 0$, then $\vec{a} \perp \vec{b}$.
- **Cross Product (Vector Product):** $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin \theta \hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$.
 - $|\vec{a} \times \vec{b}|$ = Area of parallelogram.
 - If $\vec{a} \times \vec{b} = \vec{0}$, then $\vec{a} \parallel \vec{b}$.
- **Scalar Triple Product:** $[\vec{a}, \vec{b}, \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.
 - Volume of parallelepiped.
 - If $[\vec{a}, \vec{b}, \vec{c}] = 0$, then vectors are coplanar.

Three Dimensional Geometry

- **Direction Cosines (DCs):** $\cos \alpha, \cos \beta, \cos \gamma$ where α, β, γ are angles with positive x, y, z axes.
 - $l^2 + m^2 + n^2 = 1$.
- **Direction Ratios (DRs):** Any numbers a, b, c proportional to DCs.
 - DCs from DRs: $l = \frac{a}{\sqrt{a^2+b^2+c^2}}$, etc.
- **Equation of a Line:**
 - **Passing through (x_1, y_1, z_1) with DRs a, b, c :**
 - Vector form: $\vec{r} = \vec{a} + \lambda \vec{b}$ where $\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$, $\vec{b} = a\hat{i} + b\hat{j} + c\hat{k}$.
 - Cartesian form: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$.
 - **Passing through two points (x_1, y_1, z_1) and (x_2, y_2, z_2) :**
 - Vector form: $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.
 - Cartesian form: $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$.
- **Angle between two lines:** $\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}$.
- **Shortest Distance between Skew Lines:** $\frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$.
- **Equation of a Plane:**
 - **Normal form:** $\vec{r} \cdot \hat{n} = d$.
 - **Perpendicular from origin:** $lx + my + nz = p$.
 - **Passing through (x_1, y_1, z_1) and normal vector $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$:**
 - Vector form: $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$.
 - Cartesian form: $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$.
 - **Passing through three non-collinear points:** Use determinant form.
- **Distance of a point (x_1, y_1, z_1) from a plane $Ax + By + Cz + D = 0$:** $\frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$.

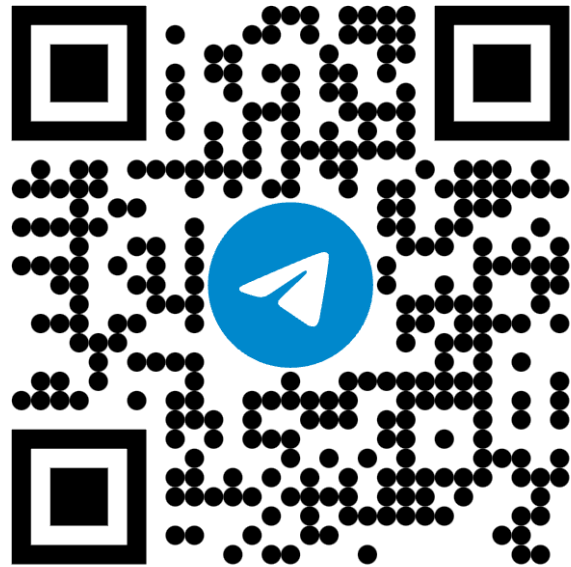
Linear Programming

- **Objective Function:** $Z = ax + by$ (to be maximized/minimized).
- **Constraints:** Linear inequalities involving decision variables x, y .
- **Feasible Region:** The common region determined by all constraints.
- **Feasible Solution:** Points within or on the boundary of the feasible region.
- **Optimal Solution:** A point in the feasible region that gives the optimal value (max/min) of the objective function.
- **Corner Point Theorem:** The optimal solution (if it exists) occurs at a corner point (vertex) of the feasible region.

Probability

- **Conditional Probability:** $P(E|F) = \frac{P(E \cap F)}{P(F)}$, where $P(F) \neq 0$.
- **Multiplication Theorem:** $P(E \cap F) = P(F)P(E|F) = P(E)P(F|E)$.
- **Independent Events:** $P(E \cap F) = P(E)P(F)$.
- **Total Probability Theorem:** If $E_1, E_2, \dots, E_n$ are mutually exclusive and exhaustive events, then $P(A) = \sum_{i=1}^n P(E_i)P(A|E_i)$.
- **Bayes' Theorem:** $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{j=1}^n P(E_j)P(A|E_j)}$.
- **Random Variable:** A real-valued function whose domain is the sample space of a random experiment.
- **Probability Distribution:** A table showing the values of a random variable and their corresponding probabilities.
 - $\sum P(X) = 1$.
- **Mean of a Random Variable ($E(X)$):** $\sum x_i P(X = x_i)$.
- **Variance of a Random Variable ($Var(X)$):** $E(X^2) - [E(X)]^2 = \sum x_i^2 P(X = x_i) - (\sum x_i P(X = x_i))^2$.
- **Bernoulli Trials:** Trials that are independent, have only two outcomes (success/failure), and probability of success is constant.
- **Binomial Distribution:** $P(X = r) = {}^n C_r p^r q^{n-r}$, where n is number of trials, p is probability of success, $q = 1 - p$ is probability of failure.
 - Mean: np .
 - Variance: npq .

@ThunderStudy



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